Aiden N. Erard

314-885-4177 | aerard3@gatech.edu | U.S. Citizen | www.linkedin.com/in/aidenerard

Education

Georgia Institute of Technology | Atlanta, GA *August 2025 – Present*
Bachelor of Science in Computer Engineering, GPA 4.0 Expected Graduation, May 2028
Minor in Computer Science, Concentration in Intelligence

Skills

Programming: Python, Java (novice), JavaScript, HTML, CSS, ROS, C++ (novice)
Hardware: Raspberry Pi, Magnetometers, Pixhawk flight controller, LiDAR/RGB-D sensors
Software: SLAM, Open3D, OpenCV, SolidWorks, GitHub
Languages: English (native), Spanish (novice)

Experience

Georgia Institute of Technology | Atlanta, Georgia *January 2026 – Present*
RoboJackets RoboNav | Drone Team Member
- Specializing in autonomous and manual drone delivery through various collegiate robotics competitions.
- Developed skills related to system integration, communication, autonomy through 8-week software training (ROS, Python, C++).

Georgia Institute of Technology | Atlanta, Georgia *January 2026 – Present*
Invention Studio | Provisional Prototyping Instructor
- Mentor students in design and prototyping at Georgia Tech's Invention Studio (largest student-run maker space in the U.S.).
- Trained/certified on equipment including 3D printers, laser cutters, CNC mills, and water jet; enforce safe operating procedures.
- Provide hands on technical guidance for class and personal projects, advising in manufacturability and troubleshooting.

Georgia Institute of Technology - LiDAR Lab| Atlanta, Georgia *January 2026 – Present*
Bipedal Navigation Over Challenging Terrain | Undergraduate Research Assistant
- Researching traversability estimations for bipedal robot navigation on rough and sloped terrain.
- Work with ROS and Python/C++ to integrate, test, and improve the navigation stack.
- Analyze simulation/real-world terrain data to evaluate stability limits and path-planning behavior.

Projects

FluxSpace | CREATE-X, Startup Exchange, Inventure Prize *September 2025 - Present*
Co-Founder
Developing magnetic-sensing drone technology to improve the safety and speed of construction/post-disaster structural inspection.
- Built a Raspberry Pi and magnetometer scanning pipeline in Python using NumPy/Pandas to collect grid-based magnetic field data, clean it, and generate 2D and 3D anomaly heatmaps via OpenCV/Open3D**.**
- Integrating autonomy and mapping groundwork through use of SLAM, RGB-D/LiDAR sensor, and Pixhawk flight-controller.
- Enduring prototype stage through CREATE-X and Startup Exchange & applying to startup incubators: Y-Combinator, a16z. Top 10% of 20,000+ Y-Combinator applicants in Winter 2026 batch.

GirdGuard | Top 5 at Georgia Tech Energy Hackathon *January 2026*
Load-shedding app targeted towards areas with unreliable electricity that lets users manage power usage from their mobile device.
- Built a live power monitoring pipeline on a Raspberry Pi and processed sensor telemetry in Python to compute live usage data.
- Developed a navigable UI using JavaScript, HTML, and CSS that displayed live readings and allowed users to add/remove loads.
- Gained hardware-software integration experience by linking dashboard controls to physical switching on the controller.

Relevant Coursework

Linear Algebra: matrix/vector operations; systems of linear equations; eigenvalues/eigenvectors.
Intro to Computing: programming fundamentals and problem solving; foundation in Python.
Intro to Object Oriented Programming: OOP principles and implementation; foundation in Java.